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Enrollment Trends at Elon University

Introduction

This study examines the factors influencing undergraduate and graduate enrollment trends at Elon University's Martha and Spencer Love School of Business from 2007 to 2024 using data mining and predictive analytics. Specifically, to explore how tuition fees, total student enrollment, and economic factors—such as GDP trends and the impact of COVID-19—have shaped enrollment patterns. Key variables include institutional factors (tuition costs, business majors offered, graduation rates, and university rankings), student demographics and enrollment trends, economic indicators, and post-graduation employment outcomes. By analyzing historical data, this study aims to identify patterns and correlations, applying machine learning models to predict future enrollment trends from 2025 to 2030. These insights will support strategic decision-making for Elon University's business school.

This study's descriptive analysis aims to explore three main areas: enrollment trends, the impact of economics and tuition, and insights specific to the business school. First, the goal is to analyze overall student enrollment trends over the years, compare undergraduate and graduate enrollment patterns, and assess enrollment growth or decline during the COVID-19 years. The second goal is to examine the correlation between tuition changes and student enrollment while also assessing GDP trends at both the national and state levels to understand their impact on enrollment shifts. Finally, we will explore specific insights to analyze how the Love School of Business enrollment respond to tuition changes and compare its enrollment trends with those of other majors at Elon University.

The predictive analysis in this study aims to forecast business school enrollment from 2025 to 2030 by identifying patterns from past trends and assessing potential risks of declining enrollment. Additionally, examine the impact of economic and institutional factors, predicting how tuition increases, or economic downturns might influence future enrollment. By simulating various economic conditions, such as recession or growth, the analysis provides insights into potential enrollment responses, aiding strategic planning for Elon University's business school.

Data Analysis and Model Implementation

The dataset used in the analysis consisted of enrollment and tuition data gathered using primary data from public sites on the Elon University webpage. Also included in the dataset was the Elon US rankings and COVID case dataset. Additional secondary data including the GDP for the USA was gathered from public data sites. Three models were implemented:

1. Multiple Linear Regression (MLR), a traditional linear regression approach
2. Random Forest Regressor, a non-linear ensemble learning model.
3. ARIMAX

This report analyzes and compares two machine learning models, Random Forest Regressor and Multiple Linear Regression (MLR), for predicting undergraduate and graduate student enrollments. The evaluation focuses on predictive accuracy, feature importance, and overall suitability for the dataset. The goal is to identify the model that delivers the most reliable enrollment predictions and to analyze the key factors influencing these predictions.

Data Overview

The dataset consists of 35 entries with 17 features related to student enrollment trends at Elon University's business school. Key features include undergraduate and graduate business school enrollment numbers, economic indicators such as GDP (US) and GDP (NC), Covid-19-related factors including Covid cases and binary indicators of Covid impact, as well as tuition fees and university rankings. During data preprocessing, it was observed that tuition fees and Covid cases were stored as text and required conversion into numerical format. Additionally, potential missing values in some fields were identified but were not explicitly detailed in the dataset summary. These preprocessing steps were essential to ensure the accuracy and effectiveness of the machine learning models.

Two predictive models were implemented: the Random Forest Regressor, a non-linear ensemble learning model, and Multiple Linear Regression (MLR), a traditional regression approach that assumes linear relationships between independent and dependent variables.

The Random Forest model was trained with optimized hyperparameters to improve performance. The undergraduate model was configured with 100 trees, a maximum depth of 15, a minimum sample split of 2, and a minimum sample leaf of 2. The graduate model was trained with 50 trees, a maximum depth of 5, a minimum sample split of 4, and a minimum sample leaf of 2. Random Forest offers key advantages, such as its ability to handle non-linearity effectively and its robustness against overfitting when properly tuned. However, its primary challenge lies in its complexity, making it less interpretable compared to linear models.

The Multiple Linear Regression model *(reference Appendix table 1)* was developed using key predictors, including NC students, international students, business school major enrollments (undergraduate and graduate), Covid cases, and university rankings. MLR is valued for its interpretability, allowing clear explanations of the impact of individual predictors. However, it assumes a strict linear relationship between variables, which makes it sensitive to multicollinearity and may not fully capture the complexities present in the dataset.

To evaluate the predictive power of both models, several performance metrics were analyzed, including R-squared (R^2), Mean Absolute Error (MAE), and Mean Squared Error (MSE). Table 2, below summarizes the performance of the models for both undergraduate and graduate enrollments:

Model	**Undergraduate Performance**	**Graduate Performance**
Random Forest	R^2 = 0.91, MAE = 56.92, MSE = 4415.05	R^2 = 0.79, MAE = 6.91, MSE = 60.13
MLR	R^2 = 0.82, MAE = 81.52, MSE = 9162.68	R^2 = 0.50, MAE = 10.88, MSE = 139.69

The results indicate that Random Forest consistently outperformed MLR across both undergraduate and graduate enrollment predictions. For undergraduate enrollment, Random Forest achieved an R^2 of 0.91, demonstrating a strong model fit, whereas MLR lagged behind at 0.82.

Additionally, the Random Forest model exhibited a lower MAE of 56.92, meaning it produced fewer prediction errors compared to MLR's 81.52.

For graduate enrollment, Random Forest continued to demonstrate superior performance, with an R^2 value of 0.79 compared to MLR's 0.50. The Random Forest model also had significantly lower error rates, with an MAE of 6.91 and an MSE of 60.13, compared to MLR's 10.88 and 139.69, respectively. These findings reinforce that Random Forest is a more reliable predictive model for enrollment trends, particularly given its ability to handle complex, non-linear relationships within the dataset.

Feature Importance & Interpretability (Random Forest Model)

Feature importance analysis was conducted to understand the key drivers of student enrollment predictions. As seen below in table 3, the most influential features for both undergraduate and graduate enrollment models:

Table 3: Feature Importance

Feature	**Undergraduate Importance**	**Graduate Importance**
Tuition Fees (year)	0.4938	0.2380
GDP (NC)	0.1763	0.3144
GDP (US)	0.1615	0.3349
Total Majors (UG/GS)	0.1367	0.0856
Business School Major Enrollment	0.0253	0.0074
Covid-19 Factors	~0.002-0.016	~0.002-0.016
Elon Ranking (USA)	0.0024	0.0014

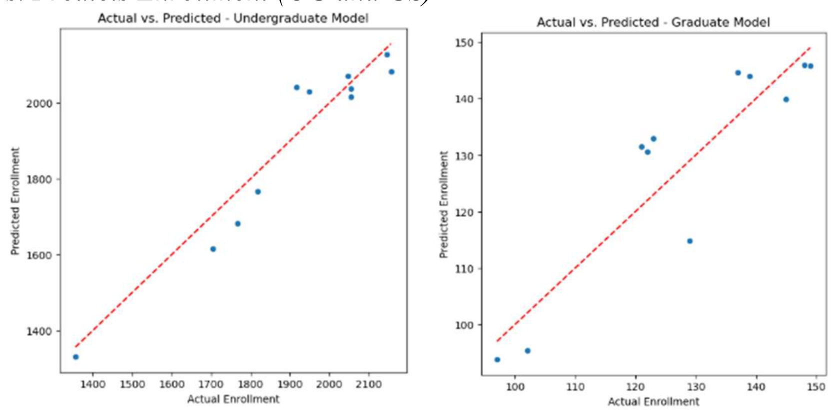
For undergraduate enrollment, tuition fees emerged as the most significant predictor, with an importance score of 0.4938, underscoring the strong influence of financial considerations on students' enrollment decisions. Macroeconomic indicators, including U.S. GDP (0.1615) and North Carolina GDP (0.1763), were also highly influential, indicating that broader economic conditions play a crucial role in students' ability to afford and commit to undergraduate education. The total number of available undergraduate majors (0.1367) was another notable factor, suggesting that program diversity may impact enrollment levels. Interestingly, variables related to COVID-19, such as the presence of the pandemic (0.0027) and COVID-19 cases (0.0013), had minimal influence, implying that undergraduate enrollment decisions were less affected by the pandemic in recent periods.

For graduate enrollment, U.S. GDP (0.3349) and North Carolina GDP (0.3144) were the two most influential predictors, reinforcing the idea that economic conditions heavily impact graduate school enrollment. Tuition fees (0.2380) also played a substantial role, further emphasizing the financial considerations in graduate education. The total number of graduate majors (0.0856) had a moderate impact, indicating that program availability may affect prospective students' choices. Similar to the undergraduate model, COVID-19 factors exhibited relatively low importance, with COVID-19 presence (0.0021) and COVID-19 cases (0.0162) playing a minimal role, suggesting that the long-term effects of the pandemic on enrollment may have diminished over time.

Visualization Interpretation and Testing

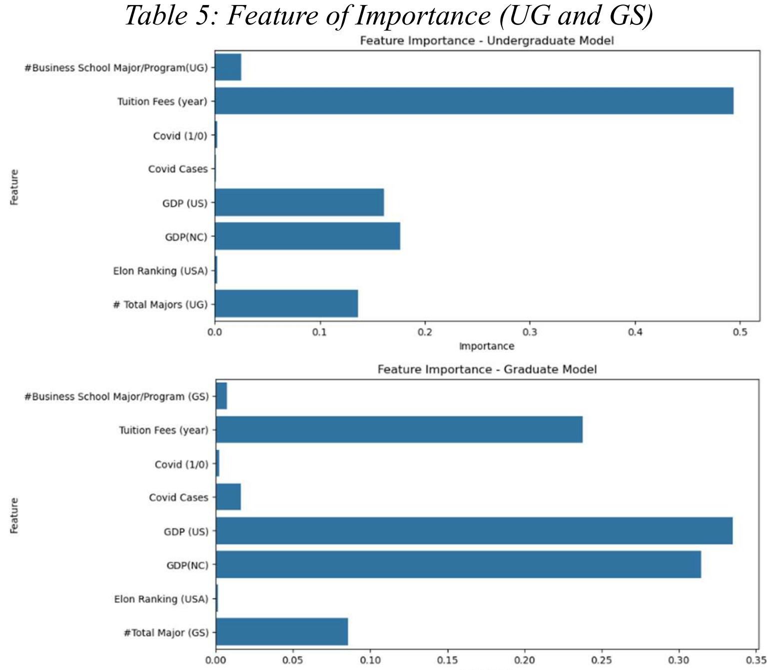
For the random forest model, the graphical summaries further support these findings by illustrating key trends in the data. As seen below in table 4, the scatter plots comparing actual versus predicted enrollment values for both models demonstrate a strong alignment, particularly in the undergraduate model, where predictions closely track actual values. This further validates the high R^2 score observed in the undergraduate model. In the graduate model, while the predictions generally follow the actual values, there is slightly greater variability, aligning with the lower R^2 value compared to the undergraduate model.

Table 4: Actual vs. Predicts Enrollment (UG and GS)



The feature importance bar charts, seen in table 5 visually highlight the dominance of tuition fees and GDP indicators in predicting enrollment. The stark contrast in importance scores between tuition fees and other features in the undergraduate model underscores its significant role, while the graduate model's more balanced distribution between tuition fees and economic indicators reflects the nuanced decision-making process in graduate education.

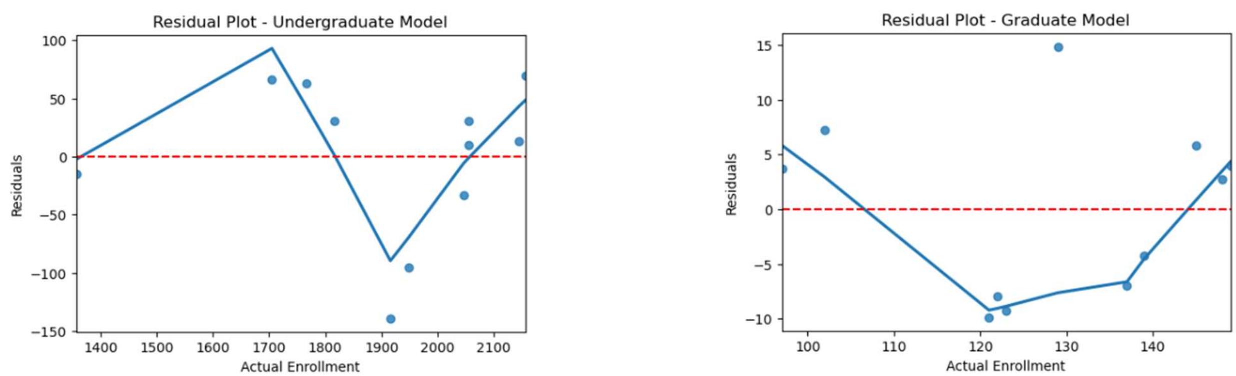
Table 5: Feature of Importance (UG and GS)



Lastly, the residual plot visualizations shown below in table 6, for the undergraduate and graduate enrollment models reveal distinct patterns that can provide deeper insights into the model's fit and potential issues. For the undergraduate model, there are noticeable fluctuations in the residuals, with values deviating significantly above and below the zero line, especially as the actual enrollment numbers increase. This suggests some inconsistencies in the model's prediction as enrollment figures grow larger, which may point to non-linear relationships or unaccounted-for variables that affect enrollment trends.

The graduate model's residual plot shows a more consistent, albeit slight, deviation from the zero line. The residuals fluctuate around zero with some larger variations, especially at lower enrollment levels. This might indicate that the model performs less accurately at these levels, potentially due to overfitting or the model not capturing specific patterns within the smaller range of graduate enrollments.

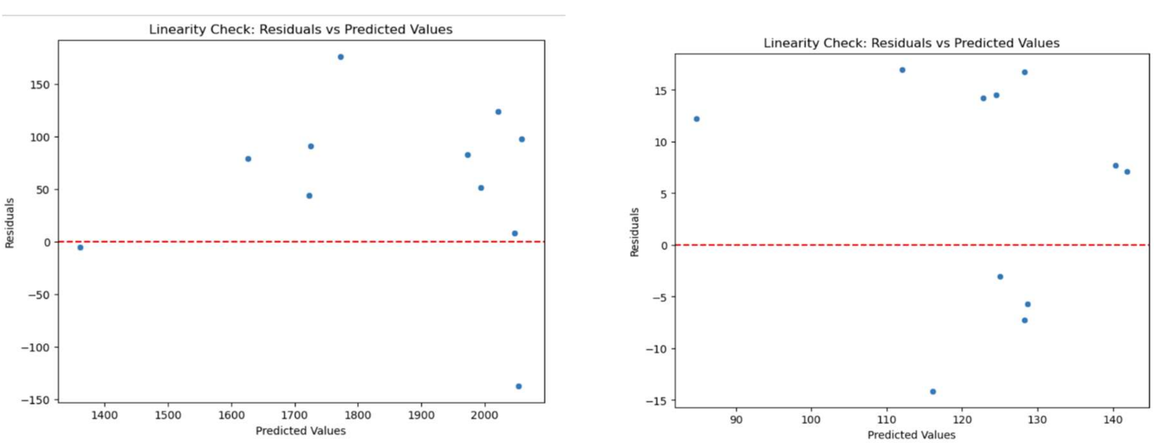
Table 6: Actual Enrollment (UG and GS)



Multiple Linear Regression Model

As seen in table 7, a linearity check was performed to validate the assumption that the relationship between the independent variables and the dependent variable follows a linear trend. Residual plot analysis showed a clear pattern, suggesting that the linearity assumption may not hold completely. Ideally, residuals should be randomly distributed with no discernible shape, but clustering and trends suggest potential non-linearity in the data seen below.

Table 7: Linearity Check



Additional statistical tests confirmed these observations. The Durbin-Watson test results indicated minimal autocorrelation, with values of 2.02 for the undergraduate model and 2.34 for the graduate model, both within acceptable ranges. However, the Breusch-Pagan test for homoscedasticity revealed that while the undergraduate model displayed constant variance (p-value = 0.45), the graduate model exhibited heteroscedasticity (p-value = 0.014), suggesting non-constant variance.

The MLR model may not be ideal for graduate enrollment predictions due to heteroscedasticity, which can lead to unreliable estimates. In contrast, the Random Forest model, free from linearity assumptions, captures non-linear patterns and interactions, making it more robust and accurate for predicting enrollment trends. While both models show some non-random residual patterns, refinement is needed. The residual plots do not directly align with economic conditions or tuition fees as expected but highlight areas for further investigation into factors influencing enrollment trends.

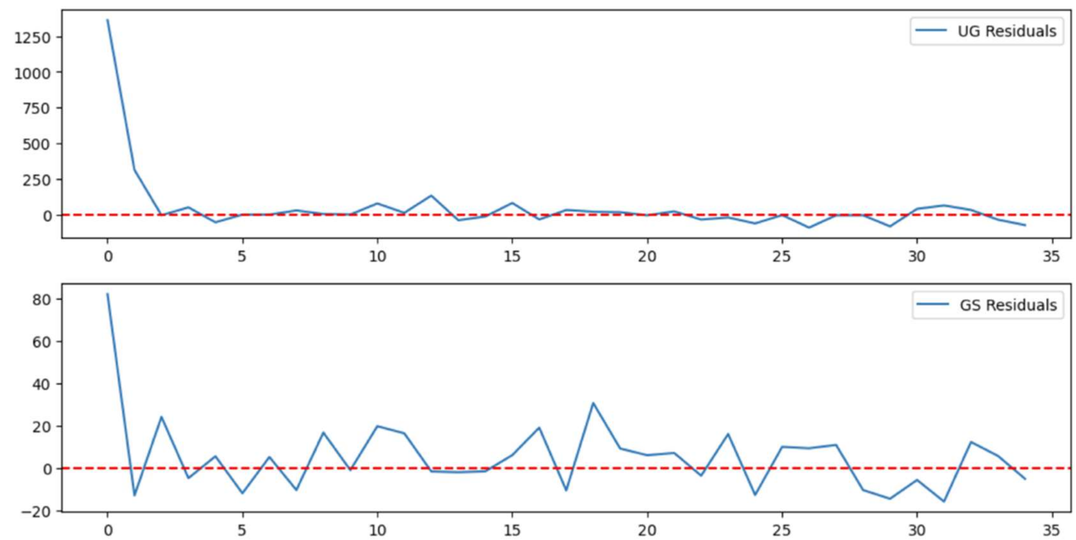
ARIMAX Model

This model is suitable for situations with complex patterns, including trends, seasonality, and the influence of external factors. We used an ARIMAX model to forecast a time series that is significantly influenced by external variables, meaning other factors besides its own past values are impacting the data, like in GDP where it might affect the enrollment figures.

To predict the impacts on enrollment and determine how tuition fees and economic factors influence total student enrollment, and economic factors we used an ARIMAX model to see how facts have shaped enrollment patterns. For undergraduate (UG) business students, the ARIMAX(2,1,2) model suggests that GDP has a strong positive impact on enrollment (coef = 234.84, p = 0.002), meaning that as GDP increases, UG student enrollment rises significantly. However, the effect of Covid-19 is positive but not statistically significant (coef = 21.15, p = 0.250), which could indicate that while the pandemic influenced enrollment, its impact was not conclusive within this dataset. The model diagnostics (AIC = 343.47, BIC = 353.51) indicate a moderate fit, and the residual tests suggest no major heteroskedasticity or autocorrelation issues.

For graduate (GS) business students, the ARIMAX(0,1,1) model reveals a negative and significant relationship between GDP and enrollment (coef = -28.66, p = 0.004), implying that higher GDP levels reduce GS enrollment—possibly because a strong economy provides more job opportunities, discouraging further education. The impact of Covid-19 is negative (coef = -5.07) but not statistically significant (p = 0.151), suggesting that while the pandemic may have discouraged graduate enrollment, the effect is not strong enough to be definitive. Model fit indicators (AIC = 256.55, BIC = 262.41) are reasonable, with no major residual issues.

Table 8: Undergraduate (UG) and Graduate (GS) Business School Students COVID Impact on Enrollment



As seen in table 8 for the undergraduate business school students, COVID-19 significantly reduced undergraduate (UG) enrollment (coef = -26.71, $p < 0.001$), while GDP had a small but positive effect (coef = 5.58e-11, $p = 0.013$), indicating enrollment increases with economic growth. The positive autoregressive term (AR1 = 0.5002, $p < 0.001$) suggests that past UG enrollment strongly influences future values, whereas the negative moving average term (MA1 = -0.2739, $p < 0.001$) indicates that past shocks impact current enrollment in the opposite direction. High heteroskedasticity ($H = 3.54$, $p = 0.05$) suggests variability in errors over time, potentially affecting prediction stability. Additionally, warnings about the covariance matrix indicate that standard errors may be unreliable.

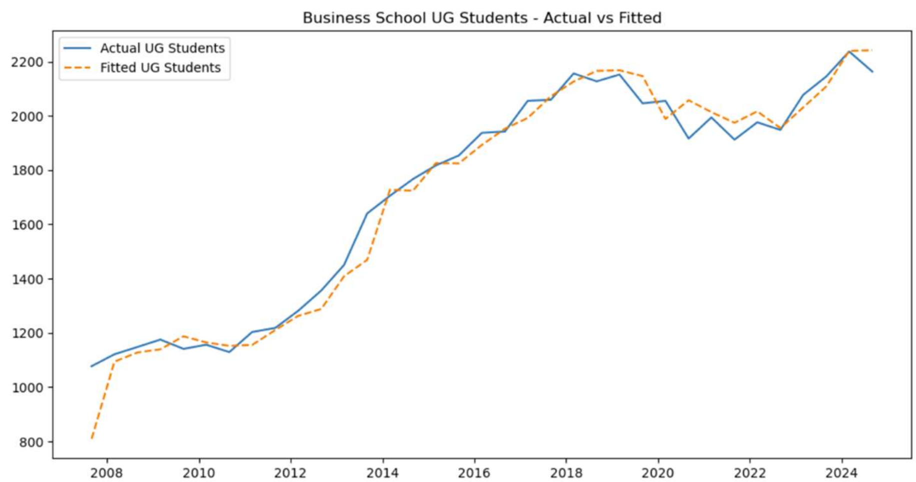
Also seen in Table 8 for the graduate business school students, COVID-19 significantly reduced graduate school (GS) enrollment (coef = -14.39, $p < 0.001$), though its impact was smaller than on undergraduate enrollment. GDP had a negative effect on GS enrollment (coef = -7.87e-12, $p = 0.003$), suggesting that economic growth reduces GS enrollment, likely due to increased job opportunities. The negative autoregressive term (AR1 = -0.6618, $p < 0.001$) indicates that past GS enrollment values inversely affect current values, while the positive moving average term (MA1 = 0.3205, $p < 0.001$) suggests past shocks have a direct positive influence. Low heteroskedasticity ($H = 0.85$, $p = 0.80$) indicates stable error variance.

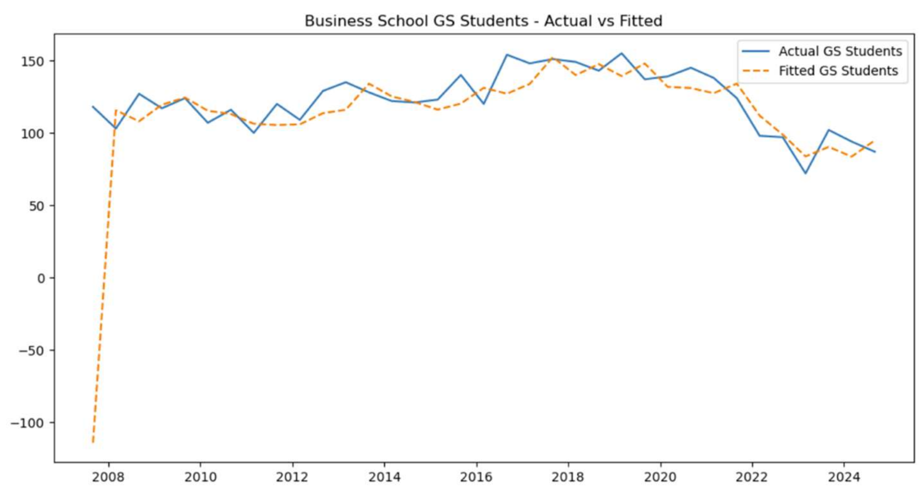
COVID-19 significantly reduced enrollment in both undergraduate (UG) and graduate school (GS) programs, with a stronger impact on UG students. Economic conditions drive opposite trends: higher GDP increases UG enrollment, likely due to greater financial security, while it decreases GS enrollment as more students opt for work over further study. Autoregressive and moving average components affect UG and GS enrollment differently, highlighting distinct enrollment dynamics. Additionally, variance stability issues in undergraduate enrollment predictions suggest potential challenges due to heteroskedasticity. To improve the models, further refinements such as testing different lag structures, considering additional external factors, or addressing heteroskedasticity could be useful.

Future Predictions

Based on the ARIMAX results in table 9, we can make the following predictions for future trends in Business School undergraduate (UG) and graduate (GS) student enrollments

Table 9: Undergraduate (UG) and Graduate (GS) Business School Students Future predictions





For undergraduate business students, COVID-19 had a significant negative impact on undergraduate (UG) enrollment, with a coefficient of -26.71, suggesting a similar decline could occur in future pandemic-like events. GDP positively influences UG enrollment (coef = 5.58e-11), indicating moderate growth if the economy continues to expand. The AR(1) coefficient (0.5002) shows that past enrollment trends moderately influence future values, allowing for some variability. For graduate business students, COVID-19 had a significant negative impact on undergraduate (UG) enrollment, with a coefficient of -26.71, suggesting a similar decline could occur in future pandemic-like events. GDP positively influences UG enrollment (coef = 5.58e-11), indicating moderate growth if the economy continues to expand. The AR(1) coefficient (0.5002) shows that past enrollment trends moderately influence future values, allowing for some variability.

Future predictions indicate that undergraduate (UG) enrollment will likely grow with rising GDP and no major disruptions, though economic downturns could negatively impact it. Graduate school (GS) enrollment may decline as job opportunities increase during economic growth, reducing the need for further study. While short-term fluctuations are expected due to ARIMA model persistence, long-term trends will mainly be influenced by macroeconomic factors like GDP growth and external shocks. GDP boosts UG enrollment but reduces GS enrollment, and the impact of COVID-19 on GS enrollment remains unclear. UG enrollment trends show greater persistence, while GS enrollment is more sensitive to immediate changes, suggesting that economic conditions significantly shape business school enrollments. Further model refinements could improve predictive accuracy.

Results and Insights

In conclusion, the findings of this study clearly indicate that Random Forest is the superior predictive model for student enrollment at Elon University's business school. Its higher accuracy and ability to handle non-linearity make it a more reliable choice than Multiple Linear Regression. While MLR offers greater interpretability, its predictive power is limited, particularly for graduate enrollment, where it achieved an R^2 of only 0.50.

For decision-making purposes, Random Forest is recommended as the preferred model when the primary goal is to maximize predictive accuracy. However, MLR remains a useful alternative when transparency and explainability are prioritized. Future improvements should include expanding the dataset to enhance model generalization and exploring alternative machine learning techniques such as Gradient Boosting or XGBoost to further refine predictive accuracy. Overall, Random Forest stands as the best model for enrollment prediction due to its robustness and superior performance across both undergraduate and graduate predictions.

Random Forest is superior for predictive accuracy, handling non-linearity, and resisting fluctuations in variance, consistently achieving higher R^2 values, which makes it ideal for enrollment predictions. On the other hand, ARIMAX is useful for long-term trend forecasting, as it can capture economic and external influences, but it is less accurate in the short term and sensitive to variance. For the highest accuracy, Random Forest is the recommended choice. However, for interpretable long-term trends, ARIMAX could be beneficial. Overall, Random Forest stands out as the better option due to its robustness and superior performance.

Appendix

Table 1: MLR Equations

For undergraduate enrollment prediction:

$Y^{\wedge}UG=2301.52-0.90(\text{NC Students})+0.59(\text{International Students})+44.46(\text{Business Major UG})+169.97(\text{Business Major GS})-0.000003(\text{Covid Cases})+0.54(\text{Elon Ranking})$ +error term

- The intercept (2301.52) represents the baseline enrollment when all predictors are zero.
- NC Students (-0.90): A negative coefficient suggests that as the number of NC students increases, undergraduate business enrollment slightly decreases.
- International Students (+0.59): A small positive impact, indicating a slight increase in enrollments with more international students.
- Business Major UG (+44.46): A strong positive influence, showing that an increase in business school majors significantly boosts undergraduate enrollment.
- Business Major GS (+169.97): The most impactful factor, implying that growth in graduate business school majors strongly correlates with undergraduate enrollment.
- Covid Cases (-0.000003): A very small negative effect, indicating a minimal relationship between Covid cases and undergraduate enrollment.
- Elon Ranking (+0.54): A slight positive correlation, meaning a higher Elon ranking increases enrollment numbers.

For graduate enrollment prediction:

$Y^{\wedge}GS=144.58-0.031(\text{NC Students})-0.035(\text{International Students})-1.36(\text{Business Major UG})+10.39(\text{Business Major GS})-0.0000003(\text{Covid Cases})+0.108(\text{Elon Ranking})$ +error term

- The intercept (144.58) represents the baseline graduate enrollment when all predictors are zero.
- NC Students (-0.031) & International Students (-0.035): Both have small negative coefficients, suggesting a weak inverse relationship with graduate enrollments.
- Business Major UG (-1.36): A small negative impact, indicating that undergraduate business school majors may slightly reduce graduate enrollments.
- Business Major GS (+10.39): A significant positive effect, showing that more graduate business school majors strongly increase graduate enrollment numbers.
- Covid Cases (-0.0000003): An almost negligible negative effect, confirming that Covid cases had minimal impact.
- Elon Ranking (+0.108): A small positive effect, indicating that better Elon rankings slightly improve graduate enrollments.

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